

Scientific Skills Exercise

Graphing Data and Evaluating Evidence

How Has the Rise in Atmospheric CO₂ Concentration Changed over Time? The blue curve in Figure 56.29 shows how the concentration of CO₂ in Earth's atmosphere has changed over a span of more than 50 years. For each year in that span, scientists collected daily CO₂ concentration data and used those data to calculate the average CO₂ level for that year. In this exercise, you'll graph average CO₂ concentrations for three ten-year periods and analyze how those concentrations have changed over time.

Data from the Study The data in the table below are average CO₂ concentrations (in parts per million, or ppm) at the Mauna Loa monitoring station for each of the indicated years.

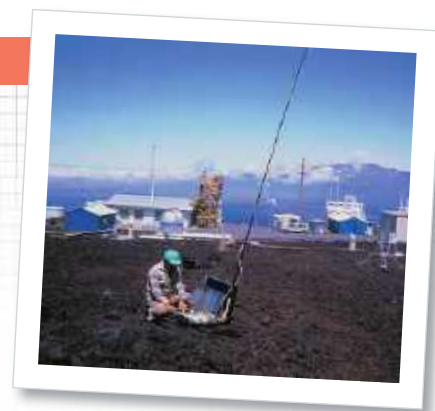
1969	324.6	1989	353.1	2009	387.4
1970	325.7	1990	354.4	2010	389.9
1971	326.3	1991	355.6	2011	391.7
1972	327.5	1992	356.5	2012	393.9
1973	329.7	1993	357.1	2013	396.5
1974	330.2	1994	358.8	2014	398.7
1975	331.1	1995	360.8	2015	400.8
1976	332.0	1996	362.6	2016	404.2
1977	333.8	1997	363.7	2017	406.6
1978	335.4	1998	366.7	2018	408.5

Data from National Oceanic & Atmospheric Administration, Earth System Research Laboratory, Global Monitoring Division.

INTERPRET THE DATA

1. Plot the data for each ten-year period on one graph, where the x-axis runs from year 1 to year 10 (producing three curves); this approach can help you to compare and contrast how the CO₂ level has changed during the three decades. Select a type of graph that is appropriate for these data, and choose

▶ A researcher sampling the air at the Mauna Loa monitoring station, Hawaii.



a vertical axis scale that allows you to clearly see how the CO₂ concentration has changed, both from year to year and from one ten-year period to the next. (For additional information about graphs, see the Scientific Skills Review in Appendix D.)

2. For each ten-year period, what is the pattern of change in CO₂ concentration? Do the curves you have drawn suggest that the pattern has changed from decade to decade?
3. Calculate the annual rate at which the CO₂ concentration increased during each ten-year period. Perform these calculations as shown here for data from 1959 to 1968 (not in the table): in 1959 the CO₂ concentration was 316 ppm, while in 1968 it was 323. Thus, from 1959 to 1968, the CO₂ concentration increased at an average annual rate of

$$\frac{323 - 316}{9}$$

or 0.8 ppm per year (we divide by 9 because during a 10-year period of time, there are 9 year-to-year changes in CO₂ concentration).

4. (a) If the annual increase in CO₂ concentration were to remain at the level observed for 2009–2018, predict the CO₂ concentration in 2100. (b) Use your results to evaluate the effectiveness of efforts to reduce CO₂ emissions.

➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

and extreme weather events (such as droughts, heat waves, and hurricanes) are occurring more often. How are such changes to the global climate affecting life on Earth?

Biological Effects of Climate Change

Many organisms, especially those such as plants that cannot disperse rapidly over long distances, may not be able to survive the rapid warming and changing precipitation patterns projected to occur over the next century. Furthermore, many habitats today are more fragmented than ever, further limiting the ability of many organisms to migrate. Indeed, the climate change that has occurred to date has *already* altered the geographic ranges of hundreds of species, in some cases leading to declining population sizes and shrinking geographic ranges (see Concept 52.1). For example, a 2015 study of 67 species of bumblebees found that on average, the geographic ranges of these important pollinators have decreased in size as the climate has warmed since 1900. Several other observed effects of climate change are discussed in **Figure 56.31**.

The ecosystems where the climate has changed the most include those in the far north, particularly northern coniferous forests and tundra. As snow and ice melt and uncover darker, more absorptive surfaces, these systems reflect less radiation back to the atmosphere and warm further (see Figure 56.30). Satellite data show that the area covered by Arctic sea ice in the summer has been declining since 1979 (the first year the data were available). At current rates of decline, there may be no summer ice there within a few decades, decreasing habitat for polar bears, seals, and seabirds. In addition, as discussed in Concept 55.2, rising temperatures have caused some Arctic regions to switch from being a CO₂ *sink* (absorbing more CO₂ from the atmosphere than they release to the atmosphere) to a CO₂ *source* (releasing more CO₂ than they absorb)—a worrisome change that could contribute to further climate warming.

Coniferous forests in western North America have also been hard hit, in this case by a combination of higher temperatures,